

Evaluation of Foundation Model Generalization to Personalized Medicine

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Abstract

Introduction

The most recent iteration of AI models (‘foundation’ and ‘world’ models) is exciting, and the field is constantly putting out newer, larger models. But the models do not generalize to out of distribution tests and do not outperform more simple models across tasks. For example, Steiner et al. [1] and Ahlmann-Eltze et al. [2] both reported that linear baselines outperformed single-cell foundation models [3,4,5,6,7] on downstream tasks using data the models had not yet seen. The Virtual Cell Challenge in 2025 found similar results on a crowd sourced model evaluation for prediction in an unseen stem cell context [8]. Translation to personalized medicine is an even more difficult goal. The prediction sets are out of distribution of the training data (i.e., transcriptional profiles from observational samples or interventions on immortalized cell lines.) Therefore, generalization is the bar these models must meet to impact personalized medicine.

This proposal creates the evaluation framework for generalization to personalized medicine. We are recruiting collaborators to contribute more tasks to test out of distribution generalization. The collective would create the biomedical AI model “acid test” [9] for precision medicine, encouraging models to demonstrate their performance on the prospective releases of test data tranches. We will pair the continual evaluation results from the [companion repository](#) with the collaborative creation of a benchmarking manuscript.

Data tranches for model evaluation

Each tranche predicts response from pre-treatment expression.

Table 1: Data tranches for evaluating foundation model generalization to personalized medicine. Each tranche predicts drug response from pre-treatment expression, with a defined held-out axis and a leakage assessment.

Tranche	Data source	Input	Output	Prediction expectation	Held-out axis	Leakage
Pre-Tranche Retrospective community data	Retrospective PDTO and cell line drug response datasets [10].	Pre-treatment expression and the compound identity.	Drug response as one fixed sensitivity metric (dose-response AUC).	Predict across cohorts, drugs, organoids/subtype, and protocols.	Held-out cohort, drug, organoid/subtype, model system.	Need to assess. Transcriptomic models may train on perturbation data from cell lines.
Tranche 1 CRC drug PDO lines	CRC drug screen: 2 cell lines and 9 patient organoids, 100 compounds, pre-treatment expression per line.	Pre-treatment expression of the line, plus compound identity.	Drug response as one fixed sensitivity metric (dose-response AUC).	Given a line’s baseline expression and a compound, predict that line’s response.	Held-out compound, organoid, drug.	Retrospective but unpublished, so unseen. Lock splits and predictions before unblinding.

Tranche	Data source	Input	Output	Prediction expectation	Held-out axis	Leakage
Tranche 2 ER+ breast, mechanism	Oliphant ER+ patient-derived xenograft organoids in BTOM-ER medium [11], with pre-treatment expression.	Pre-treatment expression, plus the ER-pathway perturbation (fulvestrant dose, or estrogen withdrawal).	Response as CellTiter-Glo viability dose-response, same metric as Tranche 1.	Predict response to the ER-pathway perturbation, and test whether the prediction tracks the ER-dependence mechanism.	Held-out mechanism: an ER-pathway or resistance state not in training. Assess explainability of predictions.	Need to assess. Existing and curated mechanism data. May leak into training.
Tranche 3 Prospective experiment	A prospective organoid experiment designed and run after predictions are locked.	Pre-treatment expression and compound identity.	Drug response as one fixed sensitivity metric (dose-response AUC).	Predict response on a not-yet-run, sealed experiment.	New patients and new contexts, tested prospectively.	Prospective and sealed. No leakage possible. Predictions registered before the assay.
Future Periodic Tranches	Prospective experiments and multi-lab datasets across multiple personalized medicine modalities.	Pre-treatment expression and the compound identity.	Perturbation response. The particular phenotype is a moving target requiring a proper evaluation adapter.	Predict response on a not-yet-run, sealed experiment or across cohorts and centers contributed by the community.	New patients and new contexts, tested prospectively, or new centers and new cohorts. Held-out mechanisms.	Prospective and sealed or unpublished.

Data. Pre-perturbation state (e.g., transcriptome). Add other modalities or clinical features in the future.

Benchmarks. Baseline linear and nonlinear statistical models on expression. Foundation and world models (STACK, STATE, X-Cell). Reported prediction performance motivates future model inclusion.

Controls. Negative: shuffled response among organoids within each drug. Positive: injected perturbation-specific effects to recapitulate.

Scoring. Perturbation by substrate interaction. For example, does the model predict that an organoid responds to a drug?

Community call. Find interested collaborators with unpublished data to assess prior to publication through social media platforms like LinkedIn. Include collaborators in Manubot-style evaluation for interpretation of results.

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The rest of this document is a full list of formatting elements/features supported by Manubot. Compare the input (`.md` files in the `/content` directory) to the output you see below.

Basic formatting

Bold text

Semi-bold text

Centered text

Right-aligned text

Italic text

Combined *italics* and **bold**

~~Strikethrough~~

1. Ordered list item
2. Ordered list item
 - a. Sub-item
 - b. Sub-item
 - i. Sub-sub-item
3. Ordered list item
 - a. Sub-item

- List item
- List item
- List item

subscript: H₂O is a liquid

superscript: 2¹⁰ is 1024.

[unicode superscripts](#)⁰¹²³⁴⁵⁶⁷⁸⁹

[unicode subscripts](#)₀₁₂₃₄₅₆₇₈₉

A long paragraph of text. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Putting each sentence on its own line has numerous benefits with regard to [editing](#) and [version control](#).

Line break without starting a new paragraph by putting two spaces at end of line.

Document organization

Document section headings:

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

A heading centered on its own printed page

Horizontal rule:

Heading 1's are recommended to be reserved for the title of the manuscript.

Heading 2's are recommended for broad sections such as *Abstract*, *Methods*, *Conclusion*, etc.

Heading 3's and Heading 4's are recommended for sub-sections.

Links

Bare URL link: <https://manubot.org>

[Long link with lots of words and stuff and junk and bleep and blah and stuff and other stuff and more stuff yeah](#)

[Link with text](#)

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[Link by reference](#)

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Multiple citations can be put inside the same set of brackets [[12](#),[16](#),[18](#)]. Manubot plugins provide easier, more convenient visualization of and navigation between citations [[13](#),[14](#),[18](#),[19](#)].

Citation tags (i.e. aliases) can be defined in their own paragraphs using Markdown's reference link syntax:

Referencing figures, tables, equations

Figure [1](#)

Figure [2](#)

Figure [3](#)

Figure [4](#)

Table [2](#)

Equation [1](#)

Equation [2](#)

Quotes and code

Quoted text

Quoted block of text

Two roads diverged in a wood, and I—
I took the one less traveled by,
And that has made all the difference.

Code `in the middle` of normal text, aka `inline code`.

Code block with Python syntax highlighting:

```
from manubot.cite.doi import expand_short_doi

def test_expand_short_doi():
    doi = expand_short_doi("10/c3bp")
    # a string too long to fit within page:
    assert doi == "10.25313/2524-2695-2018-3-vliyanie-enhansera-copia-i-
insulyatora-gypsy-na-sintez-ernk-modifikatsii-hromatina-i-
svyazyvanie-insulyatornyh-belkov-vtransfetsirovannyh-geneticheskikh-
konstruktsiyah"
```

Code block with no syntax highlighting:

```
Exporting HTML manuscript
Exporting DOCX manuscript
Exporting PDF manuscript
```

Figures

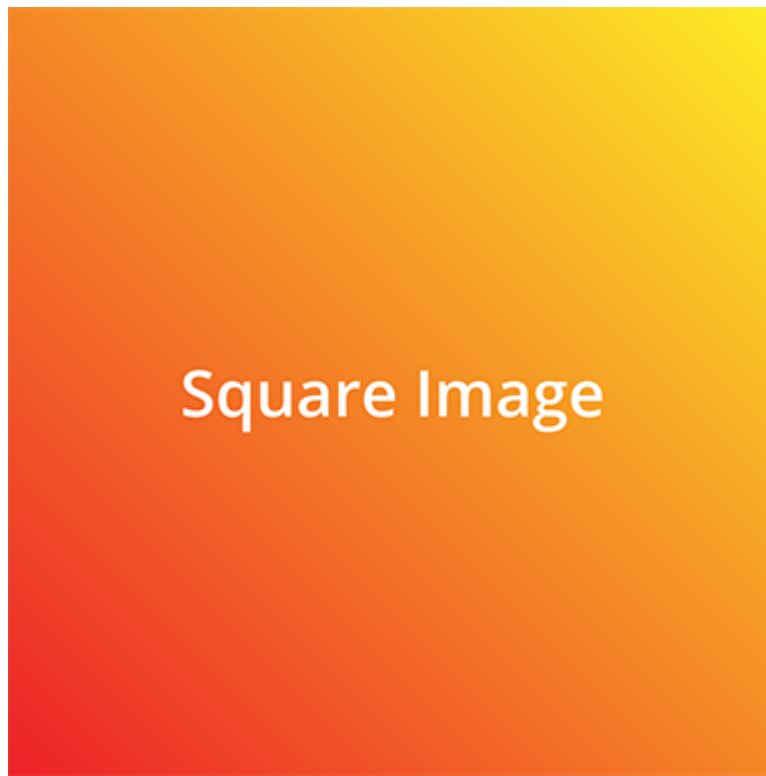


Figure 1: A square image at actual size and with a bottom caption. Loaded from the latest version of image on GitHub.

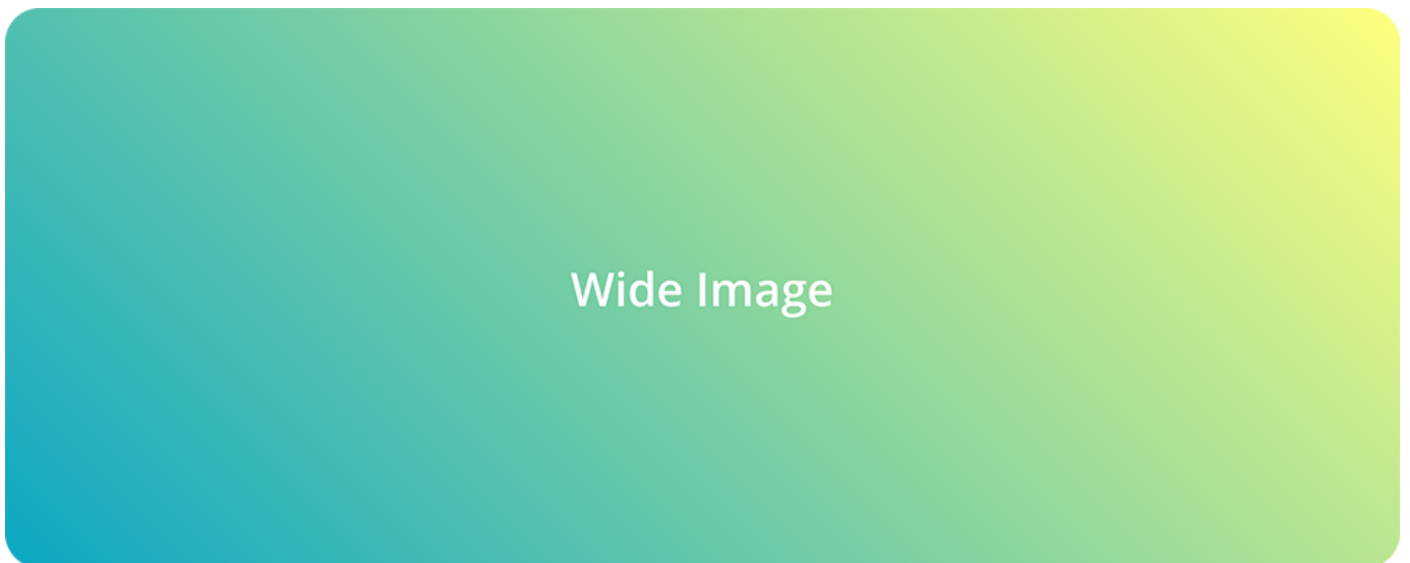


Figure 2: An image too wide to fit within page at full size. Loaded from a specific (hashed) version of the image on GitHub.

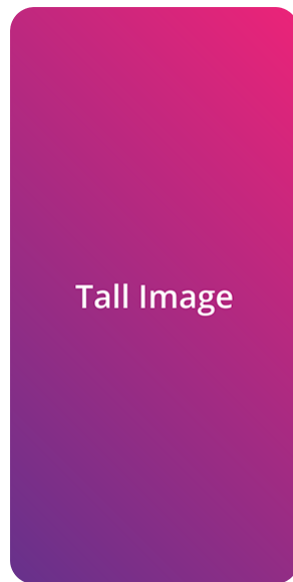


Figure 3: A tall image with a specified height. Loaded from a specific (hashed) version of the image on GitHub.



Figure 4: A vector `.svg` image loaded from GitHub. The parameter `sanitize=true` is necessary to properly load SVGs hosted via GitHub URLs. White background specified to serve as a backdrop for transparent sections of the image. Note that if you want to export to Word (`.docx`), you need to download the image and reference it locally (e.g. `content/images/vector.svg`) instead of using a URL.

Tables

Table 2: A table with a top caption and specified relative column widths.

<i>Bowling Scores</i>	Jane	John	Alice	Bob
Game 1	150	187	210	105
Game 2	98	202	197	102
Game 3	123	180	238	134

Table 3: A table too wide to fit within page.

	Digits 1-33	Digits 34-66	Digits 67-99	Ref.
pi	3.14159265358979323846264338327950	288419716939937510582097494459230	781640628620899862803482534211706	piday.org
e	2.71828182845904523536028747135266	249775724709369995957496696762772	407663035354759457138217852516642	nasa.gov

Table 4: A table with merged cells using the `attributes` plugin.

	Colors	
Size	Text Color	Background Color
big	blue	orange
small	black	white

Equations

A LaTeX equation:

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

(1)

An equation too long to fit within page:

$$x = a + b + c + d + e + f + g + h + i + j + k + l + m + n + o + p + q + r + s + t + u + v + w + x + y + z + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9$$

(2)

Special

⚠ WARNING The following features are only supported and intended for `.html` and `.pdf` exports. Journals are not likely to support them, and they may not display correctly when converted to other formats such as `.docx`.

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Adding arbitrary HTML attributes to an element using Pandoc’s attribute syntax:

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Using the [Font Awesome](#) icon set:

✓ ? ★ 🔔 ⛔ …



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